

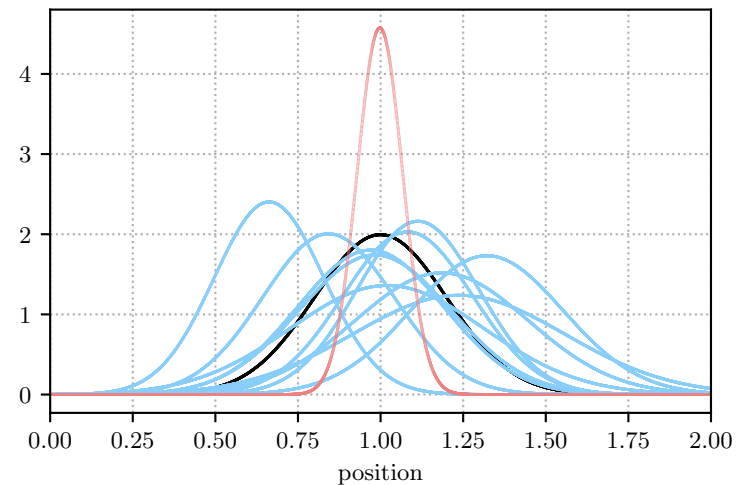
$x_i$  = data sampled from  $\text{pdf}(x|\mu, \sigma) \text{d}x = \mathcal{N}(x|\mu=1.0, \sigma=0.2) \text{d}x$  ,

$\sigma_i$  = errors sampled from  $\text{pdf}(x|\sigma, \sigma_\sigma) \text{d}x = \mathcal{N}(x|\sigma=0.2, \sigma_\sigma=0.05) \text{d}x$  :

Individual Likelihoods  $\mathcal{L}_i(\mu) = \text{pdf}(x_i|\mu, \sigma=\sigma_i) = \mathcal{N}(x_i|\mu, \sigma=\sigma_i)$  ,

$$\text{Likelihood } \mathcal{L}(\mu) = \text{pdf}(\{x_i\}|\mu, \sigma=\{\sigma_i\}) = \prod_{i=1}^{10} \mathcal{L}_i(\mu) = \prod_{i=1}^{10} \mathcal{N}(x_i|\mu, \sigma=\sigma_i)$$

Likelihood functions



Log-Likelihood functions

